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CMU-Africa

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Day reference



Night query

Problem, gap, and hypothesis

We want a robot to recognize its location at night using a daytime map. The gap we address is that many VPR methods report strong results, but it is still unclear **which descriptors transfer best to our day-to-night, campus-scale setting without retraining**. Our hypothesis is that **descriptor choice is the dominant factor** in

Descriptors evaluated

- **Older / generic:** SAD, AlexNet, NetVLAD
- **Local + global:** PatchNetVLAD
- **Modern VPR descriptors:** CosPlace, EigenPlaces, SALAD, SelaVPR++
- **Specialized spiking model:** VPRTempo

What these are

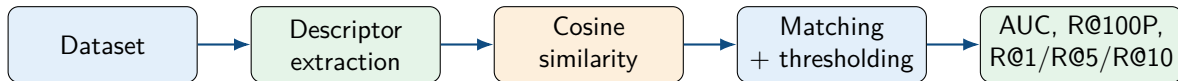
CosPlace / EigenPlaces learn strong global place descriptors. SALAD improves descriptor aggregation with DINOv2 features. SelaVPR++ adapts foundation models for efficient place recognition. VPRTempo is a fast temporally encoded spiking network rather than a universal drop-in descriptor.

Why this comparison matters

Our current phase focuses on **descriptor exploration first**: before adding CLAHE, sequence matching, or Kalman-filter-style temporal fusion, we want to know which representation is strongest in our actual setting.

Methods still planned

CLAHE preprocessing, sequence matching, and temporal prediction remain future or next-phase improvements once the descriptor baseline is settled.



GardensPoint

200 day-right reference images and 200 night-right queries. This remains our main benchmark because it directly matches the **day-to-night** problem.

Campus dataset

50 day images and 64 night queries. We evaluate it under both strict image-level labels and place-level labels for more realistic multi-view localization. All experiments here are **training-free transfer tests** unless explicitly stated otherwise.

Descriptor	AUC	R@1	R@5	R@10
CosPlace	0.592	0.350	0.840	0.925
EigenPlaces	0.676	0.370	0.840	0.940
PatchNetVLAD	0.802	0.440	0.800	0.840
SALAD	0.924	0.450	0.955	0.990
SelaVPR++	0.937	0.465	0.950	0.995
VPRTempo	0.065	0.095	0.230	0.290

Takeaway

The strongest benchmark descriptors are now **SelaVPR++** and **SALAD**. VPRTempo transfers very poorly with the official demo checkpoint.

Strict interpretation

No obvious leakage was found, but the most trustworthy

Examples: 1 correct (green) and 1 wrong (red) matches from $S \geq \text{thresh}$



Descriptor	AUC	R@100P	R@1	R@5	R@
CosPlace	0.470	0.030	0.727	1.000	1.0
HDC-DELF	0.488	0.030	0.788	0.939	0.9
SALAD	0.504	0.030	0.818	1.000	1.0
SelaVPR++	0.464	0.061	0.909	1.000	1.0
VPRTempo	0.064	0.000	0.182	0.515	0.6

Interpretation

Under strict 1-to-1 labels, **SelaVPR++ gives the strongest top-1 retrieval** while SALAD stays strong. No-match rejection remains difficult, and VPRTempo is not competitive in this transfer setting.

Examples: 3 correct (green) and 5 wrong (red) matches from $S > \text{thresh}$





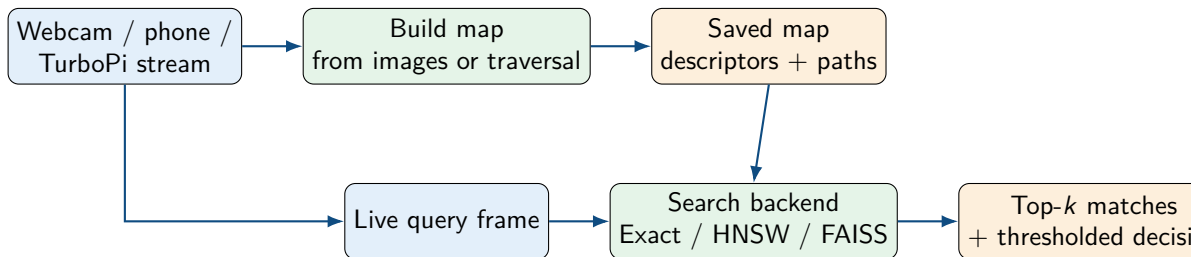
Relabeling insight

The original campus protocol was too strict for multi-view place recognition. Same-place but different-view matches were often under-credited.

Descriptor	AUC	R@100P	R@1	R@5	R@10
CosPlace	0.627	0.114	0.969	1.000	1.000
HDC-DELF	0.507	0.132	0.922	0.969	0.990
SALAD	0.653	0.130	0.969	1.000	1.000
SelaVPR++	0.658	0.158	0.969	1.000	1.000
VPRTempo	0.151	0.000	0.219	0.641	0.850

Interpretation

Once alternate views are credited properly, **CosPlace**, **SALAD**, and **SelaVPR++** all become very strong. VPRTempo improves slightly, but still transfers poorly.



Current status

We already have an end-to-end live pipeline for map building and localization. The next step is to connect the strongest descriptor choices and lightweight temporal ideas to the demo path.

- 1 Our hypothesis is supported: descriptor choice strongly affects nocturnal VPR performance.
- 2 GardensPoint remains the right primary benchmark because it directly matches the day-to-night problem.
- 3 The best descriptors we tested are now **SelaVPR++** and **SALAD**, with CosPlace still a strong practical baseline.
- 4 The campus relabeling showed that the original strict protocol under-credited valid multi-view same-place matches.
- 5 VPRTempo did **not** transfer well here, mainly because the official checkpoint is a Nordland demo model rather than a universal descriptor.
- 6 The live system is already in place, so the next phase is to connect the strongest descriptors and lightweight temporal improvements to a robust demo.

Presentation

Sunday, April 26, 2026

2:00 PM, F203

8 min presentation + 4 min Q&A

Random presentation order

Required deliverables

5-minute video presentation

Due Saturday, April 25, 11:59 PM

Final report + reproducible GitHub +
demo link